Algorithmic Pseudocode Sample 3

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 2 Initialization of the control and state variables

for $t \in \{-1, \ldots, -T^{traceback}\}$ **do** ▷Retrieve the values of $P_{u,t}$
  $P_{u,t} \leftarrow$`Power.GetValue`*(t)*
end for
for $t \in \{-1, \ldots, -T^{traceback}\}$ **do** ▷Initial conditions on the state variables
  if $P_{u,t} >$ `MinimumPower.GetValue`*(t)* **then**
    $S_{u,t}^{OFF} \leftarrow 0$
    $S_{u,t}^{STOP} \leftarrow 0$
    $S_{u,t}^{START} \leftarrow 0$
    if $P_{u,t} < P_{u,t-1}$ **then** ▷Exact initialization required only if the FLAT state is defined.
      $S_{t-1}^{ON_UP} \leftarrow 0$
      $S_{t-1}^{ON_DOWN} \leftarrow 1$
      $S_{t-1}^{ON_FLAT} \leftarrow 0$
    else if $P_{u,t} > P_{u,t-1}$ **then**
      $S_{t-1}^{ON_UP} \leftarrow 1$
      $S_{t-1}^{ON_DOWN} \leftarrow 0$
      $S_{t-1}^{ON_FLAT} \leftarrow 0$
    else if $P_{u,t} = P_{u,t-1}$ **then**
      $S_{t-1}^{ON_UP} \leftarrow 0$
      $S_{t-1}^{ON_DOWN} \leftarrow 0$
      $S_{t-1}^{ON_FLAT} \leftarrow 1$
    end if
  else if $P_{u,t} > 0$ **then** ▷Reconstruct the startups and shutdowns
    if $P_{u,t} < P_{u,t-1}$ **then**
      $S_{t}^{STOP} \leftarrow 1$
      $S_{t}^{START} \leftarrow 0$
    else
      $S_{t}^{STOP} \leftarrow 0$
      $S_{t}^{START} \leftarrow 1$
    end if
  else ▷Final possibility: the unit is OFF.
    $S_{u,t}^{OFF} \leftarrow 1$
    $S_{u,t}^{STOP} \leftarrow 0$
    $S_{u,t}^{START} \leftarrow 0$
    $S_{t-1}^{ON_UP} \leftarrow 0$
    $S_{t-1}^{ON_DOWN} \leftarrow 0$
    $S_{t-1}^{ON_FLAT} \leftarrow 0$
  end if
end for